

Genetically Informed Estimation of Marginal Returns to Education

Pietro Biroli, Elisabetta De Cao, **Tomeu López-Nieto Veitch**, Rodrigo Pinto

AIEL 2026 Conference

September 17, 2026



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA



ESSGN



Returns to College and Genetics

- Returns to college have been a long-standing focus of economic research [e.g., Becker (1964); Mincer (1974); Nybom (2017); Westphal, Kamhöfer, and Schmitz (2022)]
 - Central to productivity, mobility, and inequality
- Difficult to account for **heterogeneity in unobserved characteristics** (e.g., skills)
- **Genetics** may capture part of this heterogeneity and **interact with education** in shaping returns to college:
 - Genes may influence **selection** into education
 - **Returns** to college may **vary with genetic makeup**

Research Question

Does college **amplify** or **mitigate** genetic inequalities in earnings?

What do we contribute?

(1) Theoretical

- Extend the MTE framework to allow for **group-specific resistance to treatment**
- **Disentangle selection from genuine heterogeneity** in between-group differences in treatment effects

(2) Empirical

- Bring **genetic data** into the study of wage returns to college
- Show whether college **amplifies or mitigates earnings inequality** by genetic predisposition to education

Data

- **UK Biobank:**

- Baseline survey data collected in 2006–2010
- Genetic data, education, occupation-based earnings, and birth/current coordinates
- Cohorts born 1945–1970
- Analysis sample of about 160,000 individuals

- **Higher Education Institutions Data:**

- We construct a **new dataset** of UK higher-education institutions
- Campus locations, opening/closure dates, and university-status dates

- **Local controls:**

- pre-treatment district-level socioeconomic characteristics
- Source: 1951 Census, GBHD

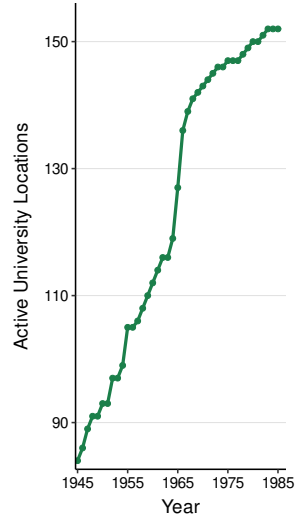
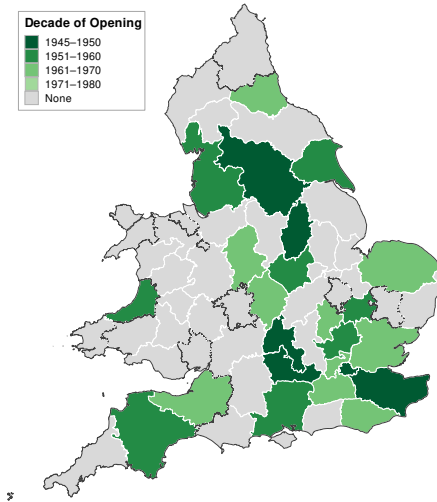
Measuring Genetic Predisposition to Education

- We use the Educational Attainment Polygenic Index (EA PGI)
- A PGI summarises the **combined association of millions of genetic variants** with a given trait (e.g., EA)
- PGIs are the **strongest available predictors** using genetic variation alone
 - The latest EA PGI explains 12–16% of variation in years of education [Okbay et al. (2022)]
- **Interpretation:** a continuous measure of genetic propensity for education
 - Fixed at conception
 - Predictive on average, but **not deterministic**
 - Not a measure of innate ability: may reflect multiple pathways

University Expansions and Distance to University

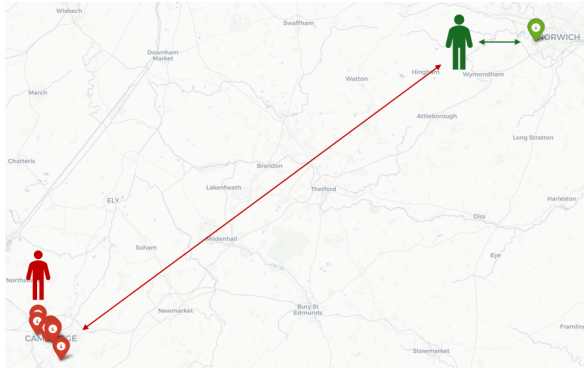
- **Distance to college** is a classic instrument for education [e.g., Card (1995); Carneiro, Heckman, and Vytlacil (2011)]
 - But it may still correlate with family background and local labour markets
- A growing literature exploits **college expansions** [e.g., Braccioli et al. (2024); Ichino, Rustichini, and Zanella (2025)]
 - To generate more plausibly exogenous variation in proximity to college
- **In this study:** we exploit the mid-20th-century expansion of universities in the UK

University Expansion in England and Wales



Who benefitted?

- Studies include **distance** to university as a **cost shifter** in pursuing education
- However, **home location determines the magnitude of the shift**



Reduction in Distance to University

(1) We measure the **reduction in distance** to the nearest university, $\Delta Dist$:

$$\Delta Dist = Dist_{prebirth} - Dist_{postbirth}$$

Summary Statistics

- *Dist_{prebirth}*: distance to nearest university, active at year of birth
- *Dist_{postbirth}*: distance to nearest university, active by age 15

(2) UK university expansion was not randomly assigned [Robbins (1963)], we control for:

- Birth-region/county and cohort fixed effects
- Local pre-treatment district characteristics: educational level, SES composition, population density/growth, unemployment, IMR, and urban/rural status.
- Initial access to higher education: number of active universities within 50km at birth

Validity of the Instrument

- **Opening Predictors and Balance**

- Openings predicted by population size and baseline provision only
- $\Delta Dist$ is balanced with respect to the EA PGI and other predetermined characteristics

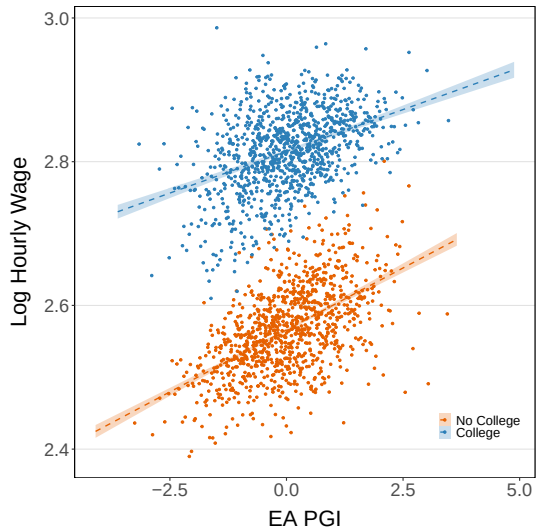
- **Placebo Exposure Tests**

- Exposure after schooling age has no detectable effect on college attendance or wages

- **First-stage relevance and functional-form sensitivity:**

- Strong first stage across fixed-effect specifications and within EA-PGI tertiles
- First-stage relevance is robust to alternative instrument definitions and functional forms

Descriptive Evidence of Gene-Environment Interplay



Benchmark Estimation of GxE

- **First Stages:**

$$E_i = \pi_0 + \pi_1 G_i + \pi_2 Z_i + \pi_3 (Z_i \times G_i) + \delta X_i + \mu_{r(i)} \alpha_{c(i)} + v_{1,i}$$
$$(G_i \times E_i) = \theta_0 + \theta_1 G_i + \theta_2 Z_i + \theta_3 (Z_i \times G_i) + \eta X_i + \mu_{r(i)} \alpha_{c(i)} + v_{2,i}$$

- **Second Stage:**

$$Y_i = \beta_0 + \beta_1 G_i + \beta_2 \hat{E}_i + \beta_3 (\widehat{G_i \times E_i}) + \gamma X_i + \mu_{r(i)} \alpha_{c(i)} + \varepsilon_i \quad (1)$$

- **Where:**

- Y : Log-hourly imputed wage
- E : College dummy
- G : Educational Attainment PGI
- Z : Reduction in Distance to university
- X : Vector of controls
- $\mu_{r(i)}, \alpha_{c(i)}$: Region of birth and cohort fixed-effects
- v_1, v_2, ε : idiosyncratic error terms

Second Stage Results

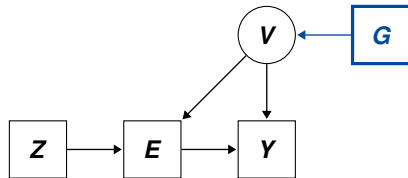
	CONTINUOUS G		DISCRETE G
	Baseline (1) Wage	County FE (2) Wage	Baseline (3) Wage
College	0.512*** (0.076)	0.588*** (0.115)	0.515*** (0.128)
EA PGI	0.034* (0.017)	0.029 (0.019)	
College × EA PGI	-0.093** (0.041)	-0.104*** (0.038)	
College × EA PGI Medium			-0.052 (0.124)
College × EA PGI High			-0.168 (0.146)
EA PGI Medium			0.020 (0.036)
EA PGI High			0.073 (0.062)
Num. Obs.	157,109	157,109	157,109
First Stage F-Stat.	19.25	5.53	12.43
R2 Adj.	0.188	0.195	0.183
Controls (X)	✓	✓	✓
Cohort FE	✓	✓	✓
Geographic FE	Region	County	Region

Standard errors in parentheses are robust to heteroskedasticity and clustered by district of birth. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. UKB weights have been applied (van Alten et al., 2024). College and its interactions are instrumented using Reduction in Distance and its interactions with G. Discrete G denotes EA-PGI tertiles, with Low as the reference category.

Why benchmark IV is not enough

- **Self-selection in returns to education** has been widely documented [e.g. Carneiro, Heckman, and Vytlačil (2011)]
- If there is self-selection that depends on G , $\widehat{\beta}_3^{\text{TSL}}$ may be **biased** [Hollenbach, Schmitz, and Westphal (2026)]
- **Solution:** Use the **Marginal Treatment Effect (MTE)** framework [Heckman and Vytlačil (2005)] to account for unobserved heterogeneity and potential self-selection in G

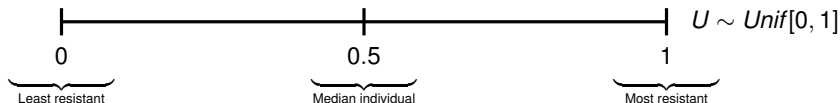
Overview of the Model



- E : College Attainment dummy
- Z : Reduction in Distance to University (IV)
- Y : Imputed Log Hourly Wage
- V : Unobserved factors affecting both E and Y (e.g. skills, motivation)
- G : EA PGI (tertiles)

Resistance to Treatment and Treatment Rule

- Following Heckman and Vytlačil (2005) we can **rank individuals** based on their **unobserved resistance to college (U)**



- and assume that an individual **attends college whenever** their **propensity, $P(Z)$** , to attend exceeds their latent resistance:

$$E = \mathbb{1}\{P(Z) \geq U\} \quad (2)$$

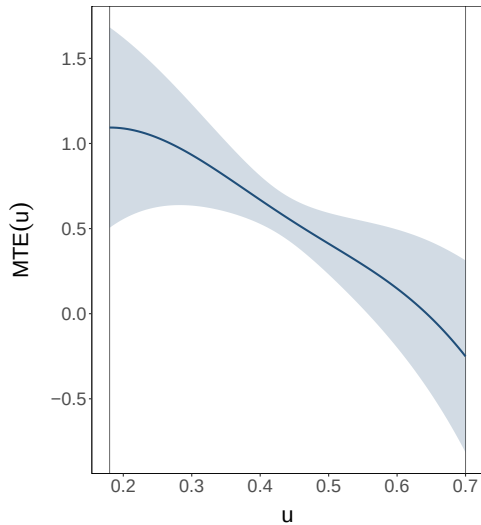
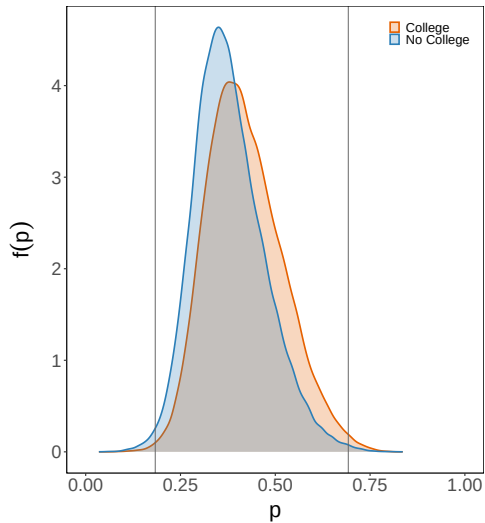
Marginal Treatment Effects

- Individuals with $U = P(Z)$ are at the **margin of indifference** between attending college and not attending
- The **Marginal Treatment Effect (MTE)** is the return to college for individuals at a given point of that margin:

$$\text{MTE}(u) \equiv \mathbb{E}(Y_1 - Y_0 \mid U = u) \quad (3)$$

- So it tells us how the returns to college vary across individuals with different latent resistance to college

Results (I): Pooled Propensity $P(Z)$ and MTE



Treatment Effects and MTE

- Following Carneiro, Heckman, and Vytlacil (2011) we can compute different treatment effects, τ , as weighted averages of the MTE:

$$\tau = \int_0^1 \text{MTE}(u) w^{\tau}(u) du \quad (4)$$

- For instance:

$$\text{ATE} = \mathbb{E}(Y_1 - Y_0) = \int_0^1 \text{MTE}(u) du$$

$$\text{ATT} = \mathbb{E}(Y_1 - Y_0 | E = 1) = \int_0^1 \text{MTE}(u) w^{\text{ATT}}(u) du$$

$$\text{ATU} = \mathbb{E}(Y_1 - Y_0 | E = 0) = \int_0^1 \text{MTE}(u) w^{\text{ATU}}(u) du$$

Results (II): Returns to college

	SUPPORT		TREATMENT EFFECTS*		
	$U_{\min}$	$U_{\max}$	ATE	ATT	ATU
Pooled MTE					
Unconditional	0.18	0.70	0.545*** [0.40, 0.70]	0.937*** [0.57, 1.30]	0.261 [-0.01, 0.53]
Conditional on G	0.05	0.90	0.530*** [0.36, 0.68]	0.967*** [0.62, 1.36]	0.235 [-0.02, 0.54]

* Approximations to the parameters, based on the empirical common support of the propensity score. Bootstrap 95% confidence intervals (1000 replicates). Conditional on G defines the MTE model that includes the EA PGI (G) as an additional observed covariate in X .

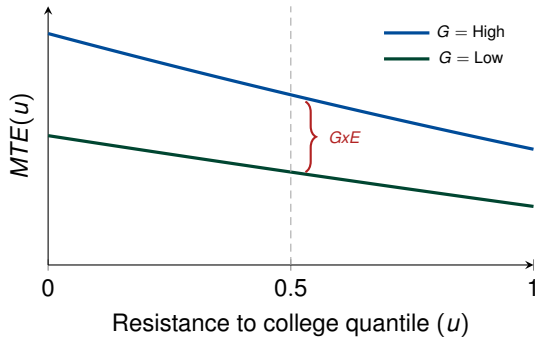
Large and Heterogeneous Returns to College

- College yields, on average, **large returns**: about 0.5 log points
 - Similar in magnitude to the raw wage gap documented by Blundell, Green, and Jin (2016)
- Yet the pooled MTE reveals **substantial heterogeneity**:
 - very large gains for those at low resistance to college
 - much smaller (possibly null) gains for those at high resistance
 - **Selection into gains**, consistent with e.g. Carneiro, Heckman, and Vytlačil (2011); Nybom (2017)

How does this heterogeneity differ by EA PGI group?

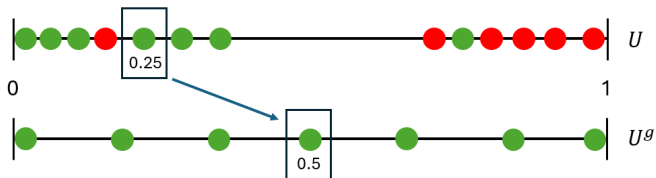
Estimating GxE via MTE

- We want to compare High PGI ($G = H$) individuals to Low PGI ($G = L$)
- Difference in MTE^g s gives the interaction effect



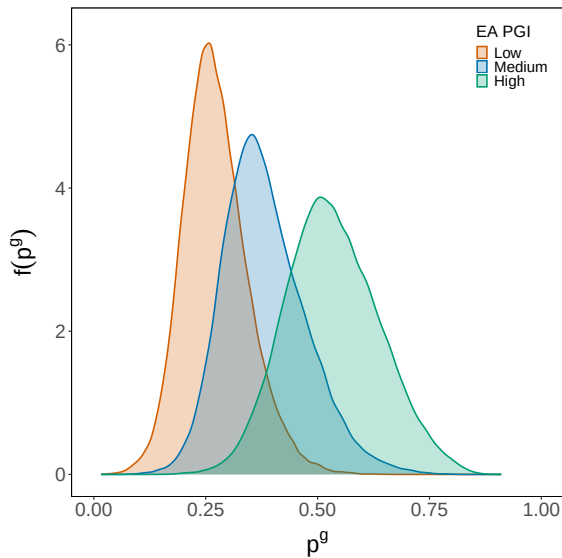
The Problem: Rank Comparability

- Is resistance universal? Most likely, $U \neq U^H \neq U^L$



- A **fixed quantile** might imply very **different latent resistance** across groups
- Differences in MTE may combine genuine heterogeneity and selection patterns

Results (III): Propensity to College by EA PGI Tertile



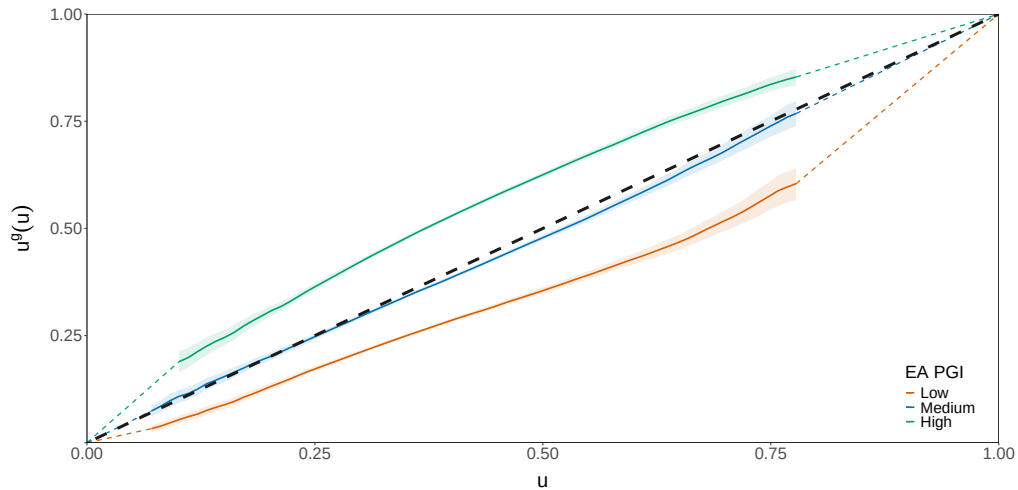
Comparing Resistances across Groups

- **Novelty:** we allow group-specific resistances (u^g) by introducing a **warp** function:

$$u^g(u) = F^g(F^{-1}(u)) : [0, 1] \rightarrow [0, 1] \quad (5)$$

- that **maps** the pooled rank u to the group-specific rank u^g
- **Intuition:** $u^g(u)$ tells us the equivalent resistance rank in group g for someone at pooled rank u

Results (IV): Estimated Relationship between Resistances



Comparing group-specific MTEs

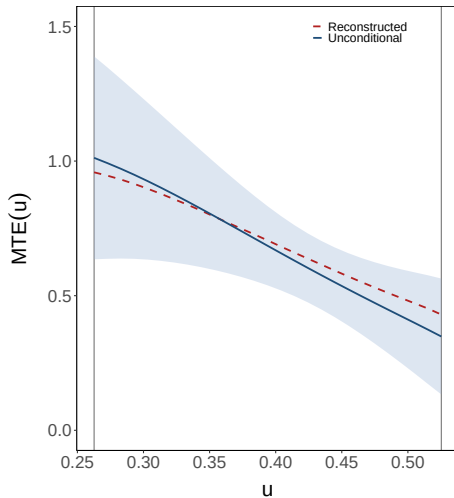
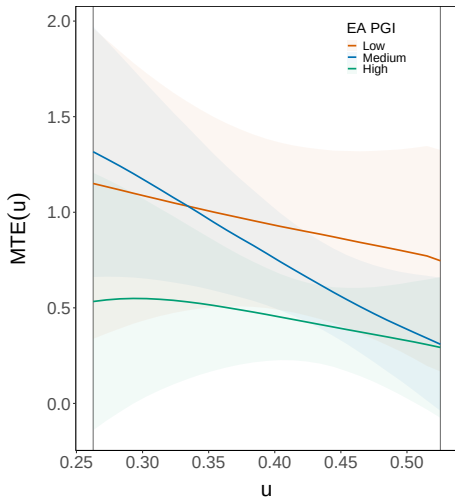
- Groups exhibit **clearly different latent resistances** to college
- To compare them, we express all MTEs on a **common resistance scale**
- We do this using the **warp** $u^g(u)$:

$$\widetilde{\text{MTE}}^g(u) = \text{MTE}^g(u^g(u)) \quad (6)$$

- The pooled MTE is then a **weighted, warped average** of group-specific MTEs:

$$\text{MTE}(u) = \sum_{g \in \mathcal{G}} \pi_g \text{MTE}^g(u^g(u)) \omega_g(u) \quad (7)$$

Results (V): Group-specific MTE



Results (VI): Returns to college by EA PGI

	SUPPORT				TREATMENT EFFECTS*		
	$u_{\min}^g$	$u_{\max}^g$	$u_{\min}$	$u_{\max}$	ATE	ATT	ATU
Group-specific MTE							
Low EA PGI	0.17	0.38	0.25	0.53	0.960*** [0.51, 1.41]	1.060*** [0.50, 1.62]	0.868*** [0.38, 1.36]
Med EA PGI	0.25	0.51	0.25	0.53	0.816*** [0.50, 1.14]	1.097*** [0.59, 1.60]	0.543*** [0.25, 0.83]
High EA PGI	0.37	0.65	0.25	0.53	0.444*** [0.17, 0.72]	0.487*** [0.04, 0.93]	0.393*** [0.14, 0.64]

*Treatment effects are evaluated over the observed common support of the resistance distribution. In the absence of full support, the reported estimates correspond to integrals over the identified region and therefore approximate the population treatment parameters defined over the full support. Entries report point estimates with bootstrap 95% confidence intervals in brackets (1000 replications).

- Group-specific results suggest **larger returns to college for Low EA PGI** individuals
- This pattern is consistent with **college mitigating inequality in earnings**

What drives these between-group differences?

Disentangling Gene-Environment Interplay

- **Two pathways of gene-environment interplay:**

- (a) EA PGI variation is associated with different selection patterns into college (rGE)

- (b) Returns to college vary depending on one's EA PGI ($G \times E$)

- **Aim:**

Disentangle PGI-linked selection into college from PGI-related heterogeneity in returns

Decomposition of GxE: Example for the ATE (I)

- We are interested in the **difference in average returns to college** between the high and low EA PGI groups ($G = H$ vs. $G = L$) in the full population:

$$\Delta^{ATE} = ATE^H - ATE^L$$
$$\Delta^{ATE} = \underbrace{\int_0^1 \widetilde{MTE}^H(u) \omega_H^{ATE}(u) du}_{ATE^H} - \underbrace{\int_0^1 \widetilde{MTE}^L(u) \omega_L^{ATE}(u) du}_{ATE^L}$$

- So the gap between high- and low-PGI groups reflects differences in both the weights and the marginal returns

Decomposition of GxE: Example for the ATE (II)

- Interestingly, the weights $\omega_G(u)$ **depend only on the propensity** to treatment
- But $\widetilde{\text{MTE}}^g(u)$ measures the **treatment effect at a given propensity** to treatment
- So that differences:
 - $\omega_H(u) - \omega_L(u)$ reflect variation in **selection patterns** rGE
 - $\widetilde{\text{MTE}}^H(u) - \widetilde{\text{MTE}}^L(u)$ reflect different **returns** at a given propensity $G \times E$

Decomposition of GxE: Example for the ATE (III)

- After some math:

Derivation

$$\Delta^{\text{ATE}} = \underbrace{\int_0^1 \left(\widetilde{\text{MTE}}^H(u) - \widetilde{\text{MTE}}^L(u) \right) \bar{\omega}(u) du}_{G \times E} + \underbrace{\int_0^1 \overline{\text{MTE}}(u) \left(\omega_H(u) - \omega_L(u) \right) du}_{\text{rGE}} \quad (8)$$

- This **approach is also applicable to other treatment effects**

See generalisation

Results (VII): Decomposition of ATE

Comparison	$G \times E$ (MTE)	rGE (Composition)	Total
Medium vs Low	-0.143 [-0.59, 0.30]	-0.006 [-0.02, 0.00]	-0.150 [-0.60, 0.30]
High vs Low	-0.505* [-0.96, -0.05]	0.007 [-0.01, 0.02]	-0.499* [-0.96, -0.04]
High vs Medium	-0.367* [-0.70, -0.03]	0.018* [0.00, 0.04]	-0.349* [-0.69, -0.01]

This table decomposes differences in average treatment effects (ATE) across EA PGI tertiles into a sorting component (rGE) and a return heterogeneity component ($G \times E$) following Proposition 3. Bootstrap 90% confidence intervals based on 1000 replications are reported in brackets.

Conclusions

- We **extended the MTE framework** to make group-specific returns comparable and decompose the differences into rGE and $G \times E$
- We find **large and heterogeneous returns to college**
- Returns are larger for **Low EA PGI** individuals, pointing to **negative** $G \times E$
- The decomposition suggests that this gap is driven mainly by $G \times E$, with only a **limited role for rGE**
- This pattern suggests that college may **reduce EA PGI-associated earnings gaps**

Thank you for listening!

`tomeu.lopeznieto@unibo.it`
`https://tomeulnv.github.io/`

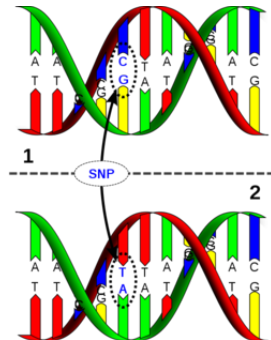


References I

- Becker, Gary S. 1964. *Human Capital: A Theoretical and Empirical Analysis, with Special Reference to Education*. New York: Columbia University Press.
- Blundell, Richard, David Green, and Wenchao (Michelle) Jin. 2016. "Big historical increase in numbers did not reduce graduates' relative wages." Institute for Fiscal Studies (IFS) Press Release. URL <https://ifs.org.uk/news/big-historical-increase-numbers-did-not-reduce-graduates-relative-wages>. Published on August 18, 2016. Accessed March 23, 2025.
- Braccioli, Federica, Paolo Ghinetti, Simone Moriconi, Costanza Naguib, and Michele Pellizzari. 2024. "Education Expansion, College Choice and Labour Market Success." Tech. Rep. 1419, Barcelona School of Economics Working Paper. BSE Working Paper No. 1419.
- Card, David. 1995. "Using Geographic Variation in College Proximity to Estimate the Return to Schooling." In *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, edited by L.N. Christofides, E.K. Grant, and R. Swidinsky. University of Toronto Press, 201–222.
- Carneiro, Pedro, James J. Heckman, and Edward J. Vytlačil. 2011. "Estimating Marginal Returns to Education." *American Economic Review* 101:2754–81. URL <http://www.aeaweb.org/articles.php?doi=10.1257/aer.101.6.2754>.
- Heckman, James J. and Edward Vytlačil. 2005. "Structural Equations, Treatment Effects, and Econometric Policy Evaluation." *Econometrica* 73 (3):669–738.
- Hollenbach, Johannes, Hendrik Schmitz, and Matthias Westphal. 2026. "Gene-environment interactions with essential heterogeneity." *The Economic Journal* :ueag010.
- Ichino, Andrea, Aldo Rustichini, and Giulio Zanella. 2025. "College, Cognitive Ability, and Socioeconomic Disadvantage: Policy Lessons from the UK in 1960–2004." *The Review of Economic Studies* 0:1–45.
- Mincer, Jacob. 1974. *Schooling, Experience, and Earnings*. New York: National Bureau of Economic Research.
- Nybom, Martin. 2017. "The Distribution of Lifetime Earnings Returns to College." *Journal of Labor Economics* 35 (4):903–952.
- Okbay, Aysu, You Wu, Ning Wang, and et al. 2022. "Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals." *Nature Genetics* 54 (4):437–449.
- Robbins, Lionel. 1963. *Higher Education: Report of the Committee Appointed by the Prime Minister Under the Chairmanship of Lord Robbins, 1961-63*. London: Her Majesty's Stationery Office (HMSO).
- Roy, A. D. 1951. "Some Thoughts on the Distribution of Earnings." *Oxford Economic Papers* 3 (2):135–146.
- The 1000 Genomes Project Consortium, A. Auton, L. D. Brooks, R. M. Durbin, E. P. Garrison, H. M. Kang, J. O. Korbel, J. L. Marchini, S. McCarthy, G. A. McVean, and G. R. Abecasis. 2015. "A global reference for human genetic variation." *Nature* 526 (7571):68–74.
- van Alten, S., B. W. Domingue, J. Faul, T. Galama, and A. T. Marees. 2024. "Reweighting UK Biobank corrects for pervasive selection bias due to volunteering." *International Journal of Epidemiology* 53 (3):dyae054. URL <https://doi.org/10.1093/ije/dyae054>.
- Westphal, Matthias, Daniel A. Kamhöfer, and Hendrik Schmitz. 2022. "Marginal College Wage Premiums Under Selection Into Employment." *The Economic Journal* 132 (646):2231–2272. URL <https://doi.org/10.1093/ej/ueac021>.

Primer in Genetics

- Human genome: series of 3 billion letter pairs (A,G,T,C)
- **Genetic variants:** one-letter changes across individuals (single nucleotide polymorphisms, SNPs)
- About 10m **SNPs** (The 1000 Genomes Project Consortium et al., 2015)
- Genome-wide association studies (**GWAS**) have identified **genome-wide significant relationships between specific SNPs and phenotypes** of all sorts.
- We **use SNPs** identified from large, replicated GWAS to **create summarised genetic scores** to study gene-by-environment interplay

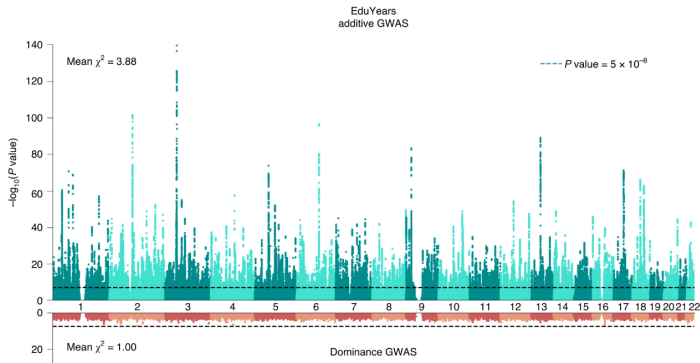


Polygenic Index (PGI)

Discovery stage:

GWAS (Okbay et al., 2022)

[Back](#)



Prediction stage:

$$PGI_i = \sum_{j=1}^J \beta_j \cdot G_{ij}$$

where G_{ij} is the genotype for individual i at SNP j , and β_j is the OLS association between SNP j and the outcome. The PGI is normalized to have mean zero and standard deviation one.

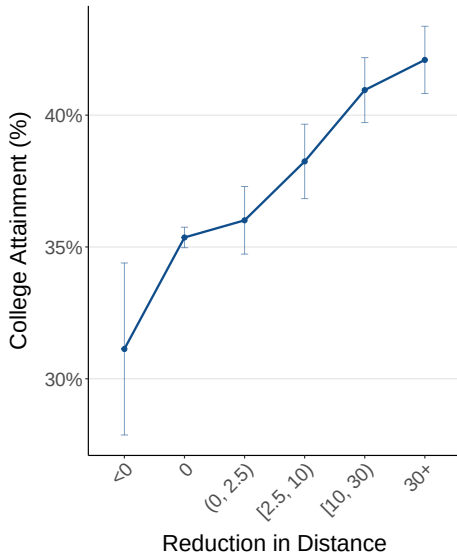
Reduction in Distance: Summary Statistics

	(1)	(2)	(3)	(4)
	Overall	Positive Reduction	Negative Reduction	No Reduction
Mean Reduction	4.721	18.949	-0.382	0
Std. Deviation	13.531	21.552	0.582	0
Min	-3.912	0.002	-3.912	0
Q1	0.000	2.606	-0.351	0
Q2 (Median)	0.000	11.928	-0.202	0
Q3	0.000	30.133	-0.135	0
Max	180.798	180.798	-0.005	0
Num. Obs.	158263	39467	1595	117201
Frequency	100%	24.93%	1.01%	74.06%

Summary statistics for the reduction in distance to university (measured in kilometers). The reduction is calculated as the difference between the distance to the nearest university active at birth and the distance to the nearest university active by age 15.

Reduction in Distance and College Attainment

Quadratic Impact Test



Test for Quadratic Impact of the Reduction in Distance

	ROBUSTNESS					
	Baseline specifications			Region $\times$ Cohort Trend		(6) EA PGI
	(1)	(2)	(3)	(4)	(5)	
	College Linear	College Quadratic	College County FE	College Linear	College Quadratic	
Reduction in Distance (Z)	0.011*** (0.001)	0.013*** (0.002)	0.006*** (0.002)	0.012*** (0.001)	0.014*** (0.002)	0.004 (0.003)
Reduction in Distance Squared (Z^2)		-0.000 (0.000)			-0.000 (0.000)	
Num. Obs.	158223	158223	158223	158223	158223	158223
R^2	0.036	0.036	0.041	0.037	0.037	0.031
Region FE	Yes	Yes	No	Yes	Yes	Yes
County FE	No	No	Yes	No	No	No
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE $\times$ Cohort Trend	No	No	No	Yes	Yes	No
Controls (X)	Yes	Yes	Yes	Yes	Yes	Yes
Genetic PCs	20	20	20	20	20	20

Notes: Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. First-Stage F-Statistic reports the F-Statistic for the excluded instrument, robust to heteroskedasticity. UKB weights have been applied. (van Alten et al., 2024). Region $\times$ cohort trend denotes a linear trend in birth cohort interacted with region-of-birth dummies. In column (3), region-of-birth fixed effects are replaced by county-of-birth fixed effects.

Timing Robustness and Placebo Tests

	ALTERNATIVE CHILDHOOD WINDOWS					PLACEBO WINDOWS			
	(1) Birth→11	(2) Birth→12	(3) Birth→13	(4) Birth→14	(5) Birth→15	(6) 20→24	(7) 24→30	(8) 30→35	(9) 35→40
Reduction in distance (Z)	0.01*** (0.000)	0.01*** (0.000)	0.01*** (0.000)	0.01*** (0.000)	0.01*** (0.000)	-0.02 (0.002)	0.07 (0.006)	0.012 (0.008)	0.000 (0.003)
Num. Obs.	158,223	158223	158223	158223	158223	158223	158223	158223	158223
R ²	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086	0.086
First-stage F	92.17	93.09	90.31	94.73	94.71	0.55	2.01	3.83	0.00
Region / Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls (X)	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

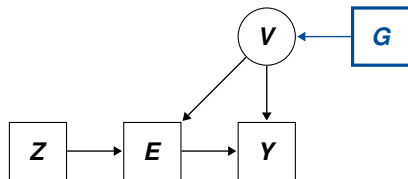
Notes: Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. First-Stage F-Statistic reports the F-Statistic for the excluded instrument, robust to heteroskedasticity. UKB weights have been applied. (van Alten et al., 2024). Region $\times$ cohort trend denotes a linear trend in birth cohort interacted with region-of-birth dummies. In column (3), region-of-birth fixed effects are replaced by county-of-birth fixed effects.

First Stage by Subgroups

	EA PGI TERTILES			SEX		SES	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Low	Medium	High	Female	Male	Low	High
Reduction in Distance (Z)	0.008*** (0.002)	0.011*** (0.002)	0.013*** (0.002)	0.012*** (0.002)	0.009*** (0.002)	0.009*** (0.002)	0.008*** (0.002)
Num. Obs.	52741	52742	52740	82865	75358	78572	79651
R2	0.023	0.034	0.043	0.037	0.039	0.030	0.036
First-stage F	19.72	29.12	40.20	65.44	33.99	22.66	39.07
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls (X)	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Outcome variable is the binary indicator for completed college degree. Robust standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. First-Stage F-Statistic reports the F-Statistic for the excluded instrument, robust to heteroskedasticity. UKB weights have been applied. (van Alten et al., 2024).

Roy Model with a Categorical Moderator



- **E**: College Attainment dummy
- **Z**: Reduction in Distance to University (IV)
- **Y**: Imputed Log Hourly Wage
- **V**: Unobserved confounder vector affecting both **E** and **Y** (e.g. skills, motivation)
- **G**: EA PGI (tertiles)

IV Model with G Setup

- Modified Roy model (Roy, 1951):
 - **Choice Equation:** $E = f_E(Z, V)$, $E \in \{0, 1\}$
 - **Outcome Equation:** $Y = f_Y(E, V, \varepsilon_Y)$
 - **Confounding Vector:** $V = f_V(G, \varepsilon_V)$
 - **Independence Condition:** $G, Z, \varepsilon_Y, \varepsilon_V$ are mutually independent
- Standard IV conditions:
 - Relevance
 - Exclusion
 - Exogeneity: $Z, G, \varepsilon_Y, \varepsilon_V$ are mutually independent
 - Monotonicity

Resistance to Treatment Ranks

- For any measurable $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$ we have that $S \equiv \psi(\mathbf{V})$ is a random variable:

$$F_S(s) = Pr(S \leq s)$$

- For individuals in group $g \in \mathcal{G}$:

$$F_S^g(s) = Pr(S \leq s \mid G = g)$$

- Which allows to **rank individuals based on their unobservables**:

$$U \equiv F_S(S) \sim Unif(0, 1)$$

$$U^g \equiv F_S^g(S) \mid G = g \sim Unif(0, 1)$$

Treatment Rules

- Following Heckman and Vytlacil (2005), an individual attains college whenever:

$$E = \mathbb{1}\{P(Z) \geq U\}, \quad E = \mathbf{1}\{P^g(Z) \geq U^g\} \text{ on } \{G = g\} \quad (9)$$

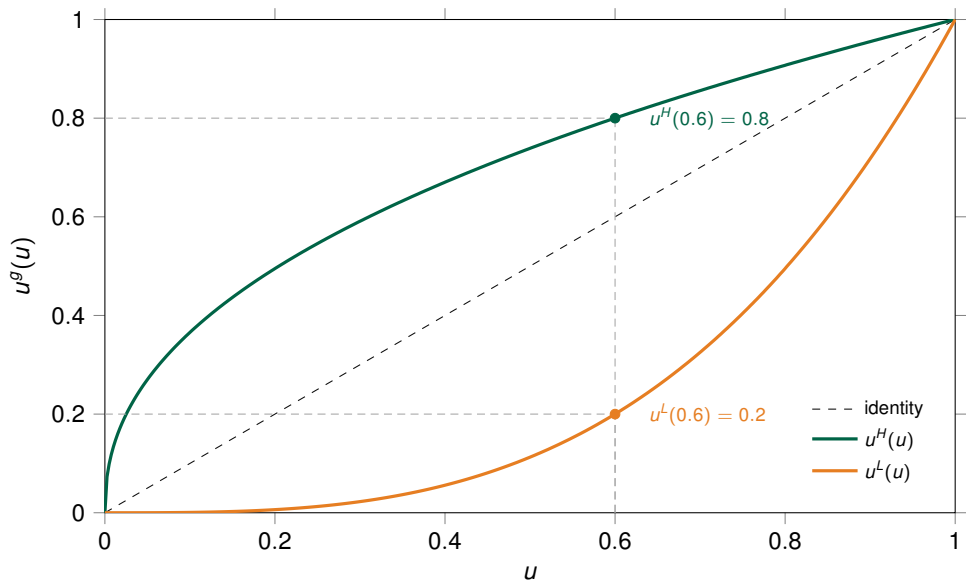
- $P(Z) \equiv P(E = 1 \mid Z)$ is the **pooled propensity score**
- $P^g(Z) \equiv P(E = 1 \mid Z, G = g)$ is the **group-specific propensity score**
- U, U^g **rank** individuals according to their **unobserved resistance to college**
 - $u \in U$: **pooled quantile** of resistance
 - $u^g \in U^g$: **group-specific quantile** of resistance
- Individuals at $P(Z) = U$ and $P^g(Z) = U^g$ on $\{G = g\}$ are indifferent

Relationship between Resistances (U , U^g)

- **Key feature:** We allow pooled and group-specific ranks **to differ**
 - Different rankings unless $F_S^g(S) \equiv F_S(S)$
- Relationship between U and U^g is governed by the **warp** $u^g : [0, 1] \rightarrow [0, 1]$:

$$u^g(u) = F_S^g(F_S^{-1}(u)) \quad (10)$$

- **Intuition:** A level u of resistance corresponds to a quantile of resistance U in the whole population, but to quantile U^g within group $G = g$
 - If $u^g < u$: Group $G = g$ is more resistant to college
 - If $u^g > u$: Group $G = g$ is less resistant to college



Properties of the warp

- By (9), the warp is identified via the **calibration identity**:

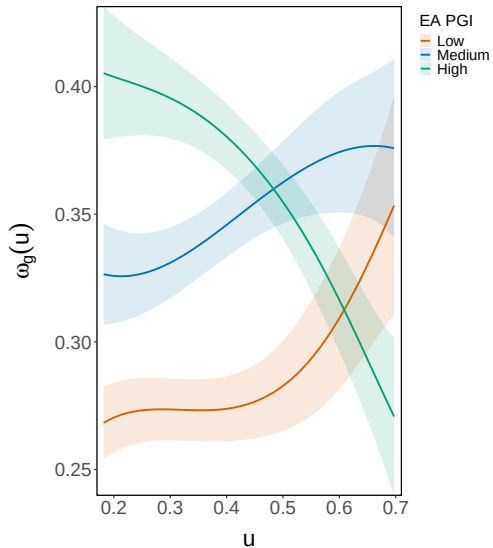
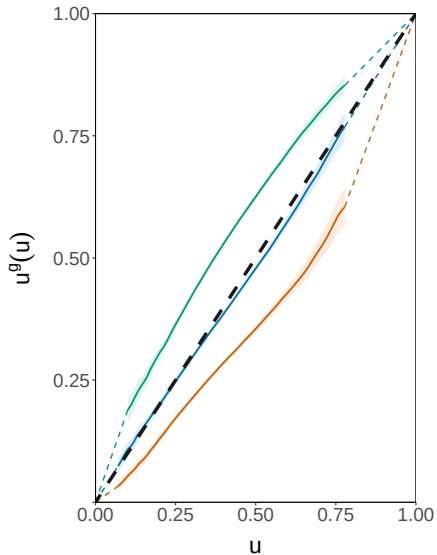
$$u^g(P(Z)) = P^g(U) \quad (11)$$

- The derivative yields the **tilt weights**:

$$\omega_g(u) = u^{g'}(u) = \frac{f_S^g(F_S^{-1}(u))}{f_S(F_S^{-1}(u))} \quad (12)$$

- “how much group g contributes to the unconditional resistance at rank u ”
- $\sum_g \pi_g \omega_g(u) = 1, \quad \forall u \in U$

Results (IV): Estimated warp and tilt weights functions



Comparing group-specific MTEs

- Groups exhibit **clearly different latent resistances** to college
- To compare them, we express all MTEs on a **common pooled resistance scale**
- We do this using the **warp** $u^g(u)$:

$$\widetilde{\text{MTE}}^g(u) = \text{MTE}^g(u^g(u)) \quad (13)$$

- The pooled MTE is then a **weighted, warped average** of group-specific MTEs:

$$\text{MTE}(u) = \sum_{g \in \mathcal{G}} \pi_g \text{MTE}^g(u^g(u)) \omega_g(u) \quad (14)$$

Decomposition of GxE: Example for the ATE (I)

- We are interested in the **difference in average returns to college** between the high and low EA PGI groups ($G = H$ vs. $G = L$) in the full population, i.e.,

$$\Delta^{\text{ATE}} = \text{ATE}^H - \text{ATE}^L$$

- The group-specific ATE is:

$$\text{ATE}^g = \int_0^1 \text{MTE}^g(u) \omega_g(u) du, \quad g \in \{H, L\} \quad (15)$$

- Hence,

$$\Delta^{\text{ATE}} = \underbrace{\int_0^1 \text{MTE}^H(u) \omega_H^{\text{ATE}}(u) du}_{\text{ATE}^H} - \underbrace{\int_0^1 \text{MTE}^L(u) \omega_L^{\text{ATE}}(u) du}_{\text{ATE}^L}$$

Decomposition of GxE: Example for the ATE (II)

- The **tilt weights** $\omega_G(u) = \frac{f_S^G(F_S^{-1}(u))}{f_S(F_S^{-1}(u))}$ depend only on selection:
 - Variation in G linked to variation in $E \implies$ rGE
- $MTE^g(u) = \mathbb{E}[Y_1 - Y_0 | U^g = u, G = g]$ measures the **treatment effect at a given resistance level**
 - Different returns at a given resistance (linked to propensity) to college $\implies$ GxE

Decomposition of GxE: Example for the ATE (III)

- Suppose there are no differences in $MTE^g(u)$ across G :

$$MTE^H(u) = MTE^L(u) = \overline{MTE}(u)$$

- Then,

$$\Delta^{ATE} = \int_0^1 \overline{MTE}(u) (\omega_H(u) - \omega_L(u)) du$$

- The difference in the ATE arise from the weights rGE

Decomposition of GxE: Example for the ATE (IV)

- Instead, suppose there are no distributional differences across G , then:

$$\omega_H = \omega_L = \bar{\omega}(u)$$

- Then,

$$\Delta^{\text{ATE}} = \int_0^1 \left(\text{MTE}^H(u) - \text{MTE}^L(u) \right) \bar{\omega}(u) du$$

- All differences in the ATE arise from the difference in the MTE $G \times E$

Decomposition of GxE: Example for the ATE (V)

- Put together:

$$\Delta^{\text{ATE}} = \underbrace{\int_0^1 \left(\text{MTE}^H(u) - \text{MTE}^L(u) \right) \bar{\omega}(u) du}_{G \times E} + \underbrace{\int_0^1 \overline{\text{MTE}}(u) \left(\omega_H(u) - \omega_L(u) \right) du}_{\text{rGE}} \quad (16)$$

- Where:

- $\bar{\omega}(u) = \frac{1}{2} \left(\omega_H(u) + \omega_L(u) \right)$
- $\overline{\text{MTE}}(u) = \frac{1}{2} \left(\text{MTE}_H(u) + \text{MTE}_L(u) \right)$

- This **approach is also applicable to other treatment effects**

[See generalisation](#)

Generalisation of the decomposition of differences in TE

- Let $A_g(u) \equiv \text{MTE}^g(u^g(u))$, $W_g(u) \equiv B_g(u) C_g(u)$
 - where $B_g(u) \equiv w_\tau^g(u^g(u))$ and $C_g(u) \equiv \omega_g(u)$

- Then, the g-specific TE is:

$$\tau^g = \int_0^1 A_g(u) B_g(u) C_g(u) du$$

- **Proposition.** The difference $\tau^g - \tau^{g'}$ admits the following exact decomposition:

$$\begin{aligned} \tau^g - \tau^{g'} &= \int_0^1 \left(\frac{W_g + W_{g'}}{2} \right) (A_g - A_{g'}) du \quad (\text{MTE channel}) \\ &+ \int_0^1 \left(\frac{A_g + A_{g'}}{2} \cdot \frac{C_g + C_{g'}}{2} \right) (B_g - B_{g'}) du \quad (\text{Target-weight channel}) \\ &+ \int_0^1 \left(\frac{A_g + A_{g'}}{2} \cdot \frac{B_g + B_{g'}}{2} \right) (C_g - C_{g'}) du \quad (\text{Composition channel}) \end{aligned}$$

